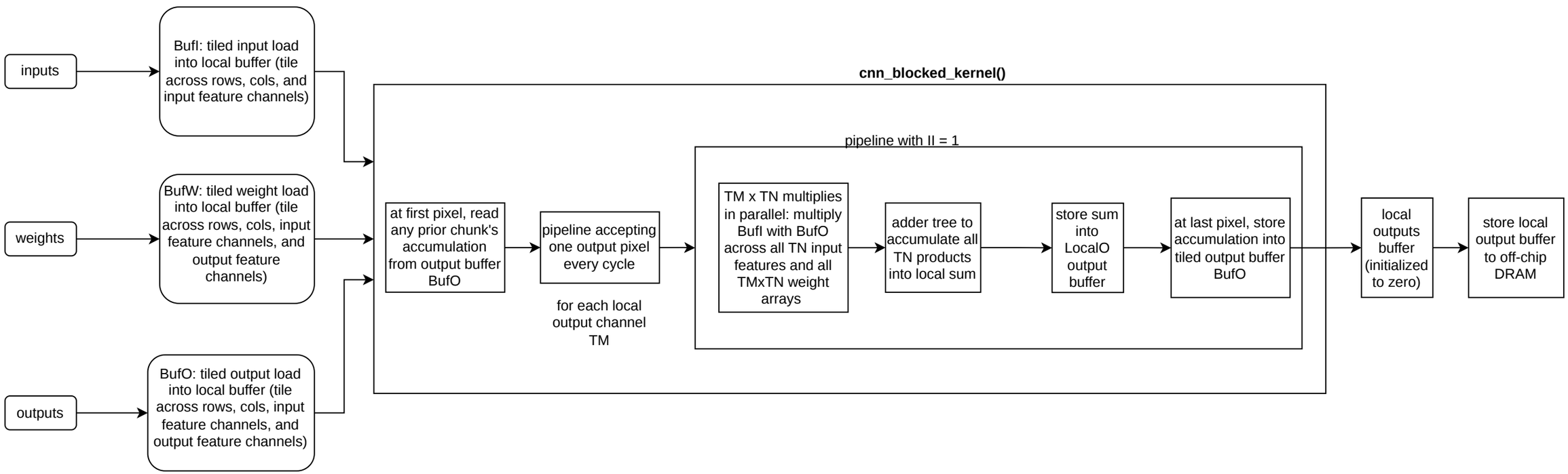
inputs
BufI: tiled input load into local buffer (tile across rows, cols, and input feature channels)
weights
BufW: tiled weight load into local buffer (tile across rows, cols, input feature channels, and output feature channels)
outputs
BufO: tiled output load into local buffer (tile across rows, cols, input feature channels, and output feature channels)
cnn_blocked_kernel()
at first pixel, read any prior chunk's accumulation from output buffer BufO
pipeline accepting one output pixel every cycle
for each local output channel TM
pipeline with II = 1
TM x TN multiplies in parallel: multiply BufI with BufO across all TN input features and all TMxTN weight arrays
adder tree to accumulate all TN products into local sum
store sum into LocalO output buffer
at last pixel, store accumulation into tiled output buffer BufO
local outputs buffer (initialized to zero)
store local output buffer to off-chip DRAM